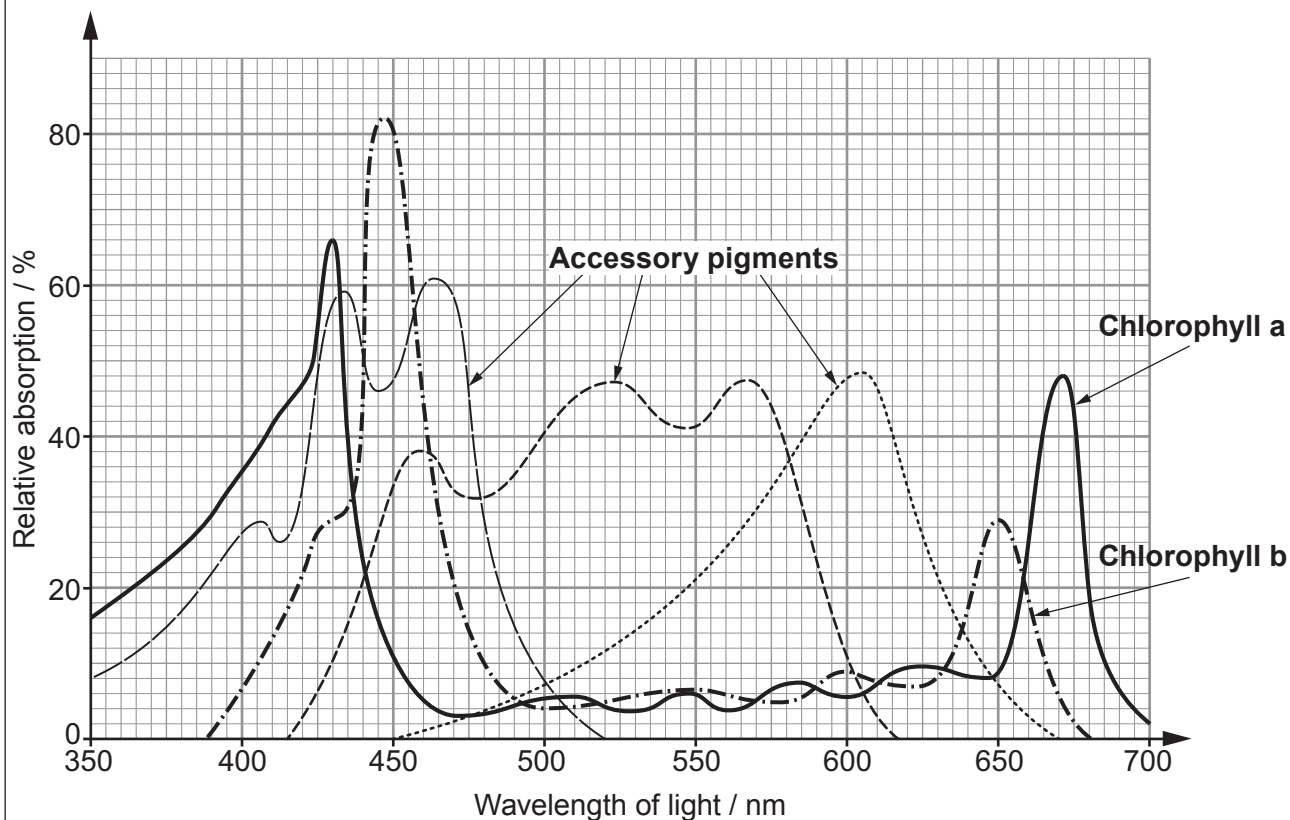


6. (a) Chlorophyll a, Chlorophyll b and other accessory pigments, absorb light energy for use in photosynthesis.

The graph below shows the absorption spectrum for each of these pigments.



- (i) Which wavelength of light is most effectively absorbed by Chlorophyll b? [1]

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- (ii) Why are accessory pigments present in the chloroplast? [1]

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- (b) Green sulfur bacteria are obligate anaerobes that are found in hot volcanic springs.

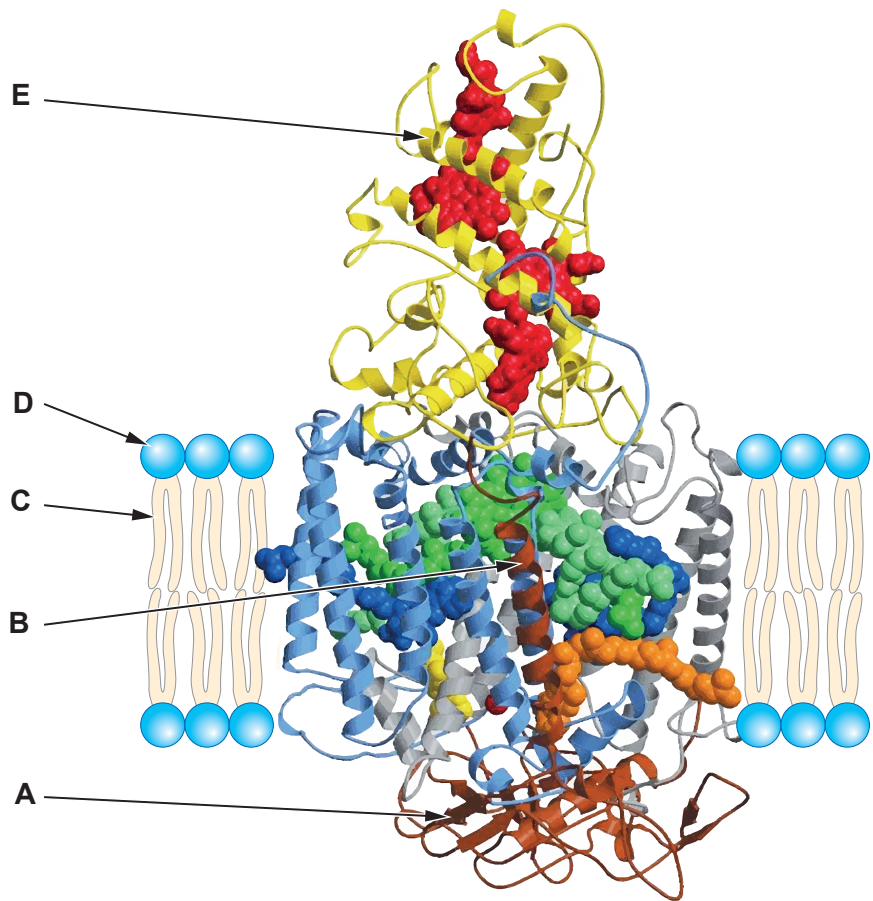
- (i) State what is meant by the term "*obligate anaerobe*". [1]

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Green sulfur bacteria are able to photosynthesise but they are different from green plants. Their plasma membranes are evolved to carry out the process.

A membrane protein complex (photosystem) contains the photosynthetic pigments. A model of this protein is shown in the diagram below.



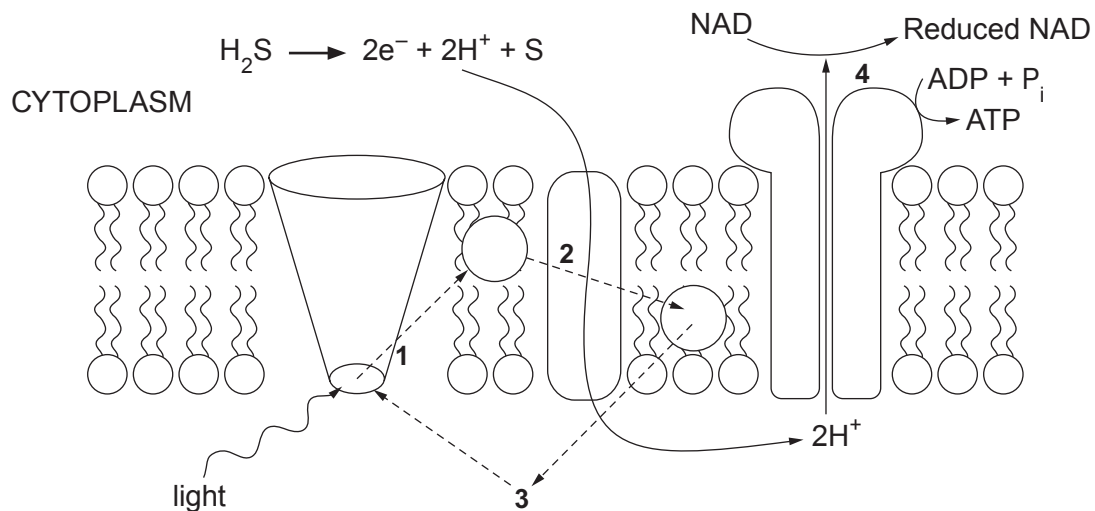
- (ii) Part of the protein is hydrophobic. Identify which letter (**A-E**) was used to label a region of the protein that would have hydrophobic properties. Explain your answer. [2]

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The diagram below shows a representation of the plasma membrane from the green sulfur bacterium *Chlorobium*. The reaction centre of this photosystem contains bacteriochlorophyll, a type of carotenoid.



1. Electrons are excited and released from the reaction centre containing bacteriochlorophyll.
 2. The electrons are passed down a series of carriers which fuel a proton pump.
 3. The electrons are passed back to the reaction centre.
 4. Protons diffuse back into the cytoplasm and reduce NAD. ATP is formed.
- (c) Using the information above and your own knowledge, describe the differences between the light-dependent reaction in green plants compared with green sulfur bacteria. [4]

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